(Theme 2 – 3135.007)

TTA: 3D-ICE-Glass: Calibrated Static & Dynamic Thermal Simulation of Photonic Glass Packages for LLM & Agentic Workloads

Benjamin Scott (PhD@UCSD-SDSU), Sohrab Sofla (PhD@PSU, CHIMES), Jason Kong (MS@UCSD), Tejinder Singh (Dell), Pranav Bussannagari (BS@UCSD), Madhavan Swaminathan (PI@PSU, CHIMES), Flavio Ponzina (PI@SDSU), Tajana Rosing (PI@UCSD)



Objectives

Problem

- Glass interposers host xPUs, stacked DRAM, RRAM accelerators and photonic interconnects in one package, beyond reticle limits
- Glass spreads heat far less than silicon: floorplan and cooling must be designed together
- LLM serving swings hundreds of watts several times a second; on-package optics drift with temperature

Limits of current practice

- **CFD (e.g., Ansys Icepak)** needs a hand-built model for every design
- **Compact solvers (HotSpot, 3D-ICE)** have no glass, Cu-TGV or optical models, and their channel correlations aren't calibrated for glass
- **Steady-state checks** give one temperature per block and miss the swings that affect optics and reliability

Contribution: 3D-ICE-Glass

- **Calibrated cooling:** every cooling option is checked once against measurement, CFD or a vendor spec
- **Optics-aware:** embedded PIC, waveguide plane and coupler are modeled with optical-exposure metrics
- **One calibrated model, two uses:** study floorplan designs and thermal dynamics under changing workloads

3D-ICE-Glass is accurate, fast & simulates thermal dynamics of complex GlassML designs:

2.6% error
for eight measured & simulated CFD baselines

0.7 K error
in microchannel peak temp

up to
975× faster
vs. SOTA CFD Icepak at comparable accuracy

3D-ICE-Glass Framework

3D-ICE-Glass is a calibrated thermal design tool for glass photonic packages, running on the standard 3D-ICE 4.0 solver for both static and dynamic analysis

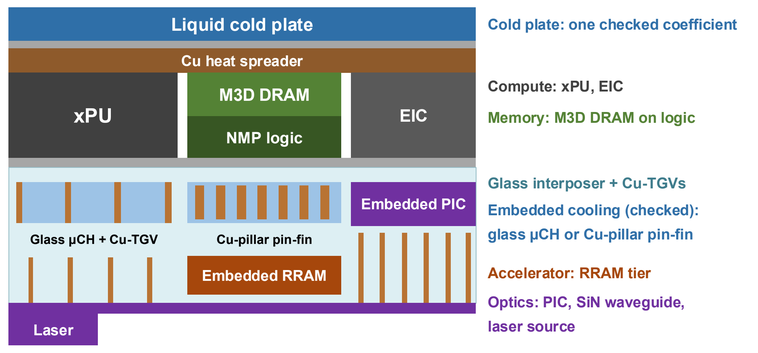


Fig. 1. What 3D-ICE-Glass models (composite of (a)–(d), not to scale). Cooling options are checked against CFD or measurement; the full package is a model prediction.

STAGE 1 Anchor once per cooling primitive

Reference data
Measurement
Conjugate CFD
Vendor spec

→ **Liquid cold plate** | **Glass µCH + Cu-TGV** | **Cu-pillar pin-fin**

Fit once at a reference point → freeze → test on a held-out design

↓ *frozen primitives*

STAGE 2 Explore designs (no CFD in the loop)

Inputs
Glass stack, Cu-TGV, embedded PIC
Chiplet floorplans
Steady power maps or simulated 5 ms serving traces

→ **3D-ICE 4.0**
Thermal RC network + coolant-channel cells
Factor once per floorplan, then sub-second transient steps

→ **Outputs**
Static: peaks vs 85 °C, headroom, sustainable power
Dynamic: PIC swing, heating rate, time above steady state

Fig. 2. 3D-ICE-Glass flow

Calibration & Validation

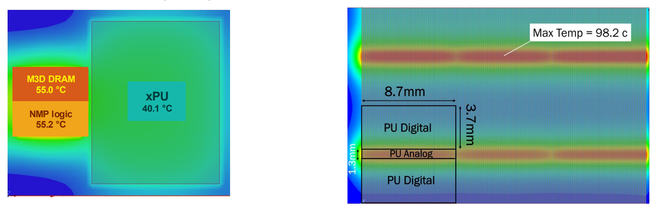
Each cooling option (liquid cold plate, coolant channels etched into the glass, copper pillars in coolant) is checked once against measurement, CFD, or a vendor spec, then frozen and reused

Table I. Validation checkpoints († bound or range check, not in error metric)

Validation case	Reference source	Role	3D-ICE-Glass	Reference	Δ
H100 test vehicle, 2-phase cold plate	Measured [InterPACK'24]	Held-out	64.8 °C	64.9 °C	**−0.1 K**
H200 cold plate†	Vendor spec (ToneCooling)	Bound	61.7 °C	≤65 °C	**−3.3 K**
Stratum-L H100 cold plate	AI model (PINN) [ECTC'26]	Cross-check	81.0 °C	86 °C	**−5.0 K**
Stratum-S H200 cold plate	Ansys Icepak (CFD)	Cross-check	55.5 °C	55.2 °C	**+0.3 K**
Glass µCH + Cu-TGV	Measured [ESTC'26]	Independent	86.4 °C	83.3 °C	**+3.1 K**
Cu-pillar pin-fin	Measured [ESTC'26]	Independent	77.6 °C	78.6 °C	**−1.0 K**
Si interlayer µCH	CFD [ITherm'23]	Flow-fit	86.5 °C	84.7 °C	**+1.8 K**
RRAM accelerator, pin-fin	Ansys Icepak (CFD)	Independent	52.2 °C	55.8 °C	**−3.6 K**
RRAM accelerator, µCH + Cu-TGV	Ansys Icepak (CFD)	Anchor	97.5 °C	98.2 °C	**−0.7 K**
Photonic PIC map†	AI model (PINN) [ECTC'26]	Range	41.8 °C	~40 °C	**+1.8 K**

Mean error: 2.6% (≈2.0 K) over 8 scored checkpoints; 3.0% (2.2 K) over the 6 not fitted to their own reference

Icepak (CFD) fields, with 3D-ICE-Glass peaks



(a) **Stratum-S [MICRO'25]**, 39.6 × 31.8 mm, liquid cold plate
Icepak 55.2 °C vs 3D-ICE-Glass 55.5 °C

(b) **HyFlexPIM [ISCA'25]**, 26.1 × 17.4 mm, µCH + Cu-TGV
Icepak 98.2 °C vs 3D-ICE-Glass 97.5 °C

Fig. 3. GlassML static chips, Icepak (CFD) temperature fields; (a) independent check, 0.3 K; (b) calibration anchor, 0.7 K. Color: blue to red ≈ 31–55 °C (a), 27–98 °C (b). Stratum-S solve time: Icepak 250 s (4 cores), 3D-ICE-Glass 70 s (12 threads; 3.6×).

Experimental Setup

GlassML static study: Stratum & HyFlexPIM on glass; steady state

- **Stratum-S [MICRO'25]:** A6000-class xPU + one 32 GB M3D DRAM stack on 39.6 × 31.8 mm, 380 µm glass; the DRAM column runs at 120 W/cm² vs 37 W/cm² in the xPU
- **HyFlexPIM [ISCA'25] RRAM AI accelerator:** 141 W per chiplet (one of four) on 454 mm², 600 µm glass; analog strips reach 190 W/cm²
- **Floorplans (a)–(c), (a) in Fig. 4:** same 39.6 × 31.8 mm, 380 µm glass under a top liquid plate; (b) adds an air-cooled spreader under the glass; (c) adds an embedded RRAM capacity tier (≈480 µm glass) with in-glass coolant

GlassML dynamic test: 89.8 × 24.4 mm glass row; transient

- **Floorplan (d), Fig. 4:** one xPU prefill node + two decode nodes (four M3D DRAM tiles on logic each), PICs embedded beneath the logic, SiN waveguide; 260 µm interposer + 20 µm waveguide plane + 100 µm PIC tier

Workload: GLM-4.5 MoE serving

- **Static:** decode-phase power per node: floor 310 W, average 398 W, TDP 519 W
- **Dynamic:** 10 s of serving (64K context, ~800 concurrent sessions, blocking KV reloads) after a 20 s warm-up; includes KV-reload idle down to ~6 W

Cooling

- **Top cold plate:** one uniform boundary (one coefficient, one coolant temperature); 27 °C water reference; 35 °C warm loop, 30% propylene glycol (plate 0.735×–0.575× of water); 45 °C = +10 K shift
- **Coolant warm-up:** not modeled along the plate; 35–45 °C covers a uniform 10 K rise. In-glass channels model coolant flow

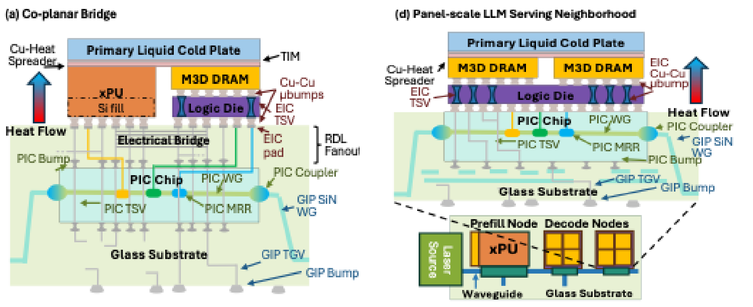
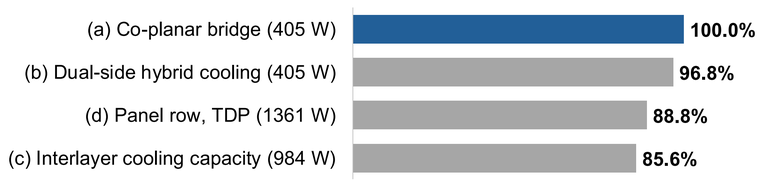


Fig. 4. GlassML floorplans: (a) static, co-planar bridge; (d) dynamic, panel-scale serving row

Results and Comparison to the State-of-the-Art

Sustained power at 35 °C inlet, conservative coolant endpoint (% of nominal)



- **Fig. 5.** At a 35 °C warm loop, only the co-planar bridge runs at nominal power without exceeding 85 °C (13 K headroom); the others derate 3–14%, though the interlayer package still delivers 842 W, 2.1× the bridge

Memory tier, panel row at TDP, 35 °C, conservative coolant (0.575× plate)

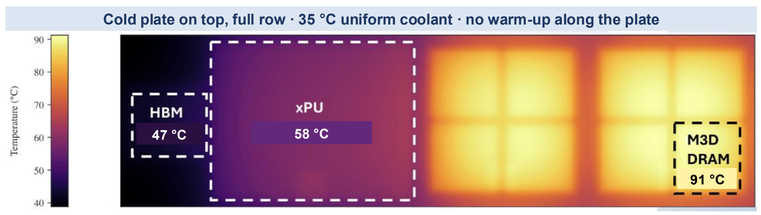
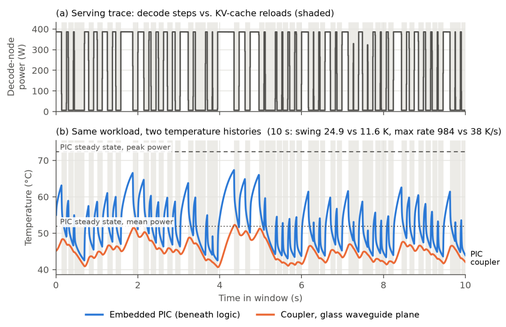


Fig. 6. Under a uniform top cold plate, memory, not the xPU, limits the row: DRAM reaches ≈91 °C at TDP (limit 85 °C); PIC spread grows from 6 K (trace floor) to 10 K (TDP)

Embedded PIC vs coupler, 10 s simulated GLM-4.5 trace (0.735× plate)



25 K
PIC swing vs 12 K at the coupler

16×
PIC heating rate vs what ring control can track

46%
of time above mean-power steady state

- **Fig. 7.** All electronics stay under 85 °C (PIC ≤ 67 °C), yet the PIC swings 25 K (~2 nm ring drift [Nanophotonics'14]). Hottest DRAM tile: 66.1 °C under serving vs 82.6 °C at TDP steady state (same 0.735× plate) (16.5 K); at the trace's own peak power, steady state overestimates by 5–6 K

Key Accomplishments, Showstoppers & Next Steps

- **Key accomplishments:**
  - **Designed 3D-ICE-Glass,** a thermal dynamics simulator for glass packages that models xPUs, stacked DRAM, RRAM accelerators, embedded photonics and Cu-TGVs with liquid cold plate, in-glass microchannel and pin-fin cooling at accuracy comparable to SOTA CDFs such as Icepak, but up to 975x faster
  - **Under serving traffic, the embedded PIC swings 25 K** and heats up to 16× faster than ring control can track, while all electronics stay under 85 °C. Steady-state sign-off misses this; TDP sign-off overestimates the DRAM peak by 16.5 K (5–6 K at the trace's own peak power)
  - **Compared four glass floorplans:** only the co-planar bridge runs at nominal power without exceeding 85 °C across the 35–45 °C warm loop; in the panel row, memory, not the xPU, is the limit (DRAM 91 °C at TDP)
- **Grand Challenge Applications:** Deep Insights
- **Demo:** static floorplan sweep + live serving-traces
  github.com/bmscott73/3D-ICE-Glass
- **Lessons learned:**
  - **Local flux, not die power, sets the limit:** the 121 mm² DRAM column runs 12 K hotter than the 628 mm² xPU
  - **A thermal pass is not optical stability:** all electronics stayed under 85 °C while the PIC swung 25 K
  - **Glass makes each die a thermal island:** the PIC follows each serving event; the waveguide plane follows average load
- **Showstoppers:**
  - **Calibration costs:** each new cooling option needs validation via measurement or CFD simulation
  - **No pump check:** the tool reports temperature, not the coolant pressure a design needs (up to 4 bar)
  - **GlassML at scale:** simulating multiple GlassML panels together requires distributed-memory parallelization of 3D-ICE
- **Next steps:**
  - **Validate and time transients** against Icepak and the PSU PINN
  - **Scale to 64-GPU and multi-panel systems:** coarse mesh away from hotspots, step-response superposition, reuse of repeated tiles
  - **Model coolant warm-up** along the plate so floorplans account for flow direction
  - **Workload-aware ring tuning:** size ring-heater control from predicted PIC swing
  - **More workloads:** long-document prefill, agent tool pauses, steady chat

This work was supported in part by the Semiconductor Research Corporation (SRC) and DARPA.

Publications:

B. Scott, S. Sofla, P. Bussannagari, T. Singh, F. Ponzina, M. Swaminathan, T. Rosing, "3D-ICE-Glass: Calibrated Thermal-Aware Floorplanning of Photonic Glass Packages for Scalable AI," DATE'27 (under review); SRC Publication #496990
B. Scott, J. Kong, P. Bussannagari, T. Singh, F. Ponzina, T. Rosing, "Trace-Driven Thermal Analysis of Glass Photonic Packages for LLM Serving," ITherm'27 (submitted); SRC Publication #496991

AVAILABLE FOR INTERNSHIP

Semiconductor Research Corporation

PRISM Annual Review | November 04-05, 2026

UC San Diego · CMU · Georgia Tech · TEXAS The University of Texas at Austin · University of Virginia · UCI · University of Illinois Urbana-Champaign · Wisconsin University of Wisconsin-Madison · UCLA · University of Washington · PennState · UTD · Stanford University · Berkeley University of California

PRISM